DLD - Neso

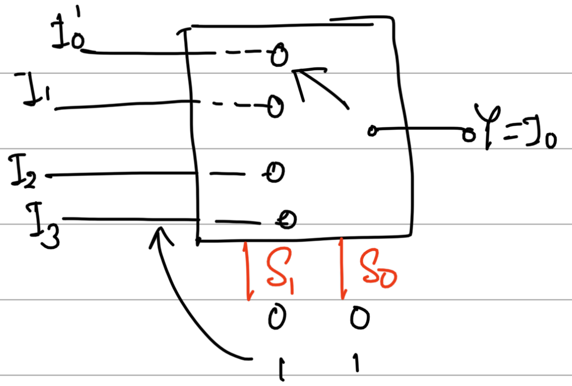
MUX

V-118:- Multiplexer

Multiplexer → combinational circuit
↳ that selects binary information from one of many input lines and directs it to o/p line.

Data selector

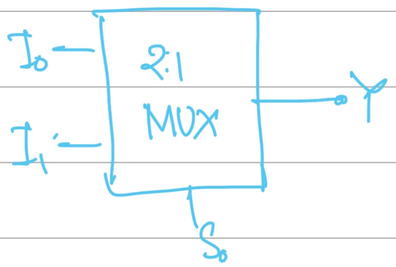
O/P always 1



Select line/selector var.

Advantage

1. Reduce no of wire
2. reduce circuit complexity
3. Implementation of various circuit using MUX



$n = 2^m$ → no. of select line

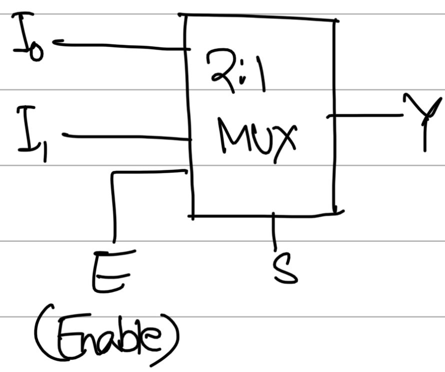
n → no. of I/P

$\boxed{m = \log_2 n}$ ⟹ $n = 4$, $m = 2$

Types:-

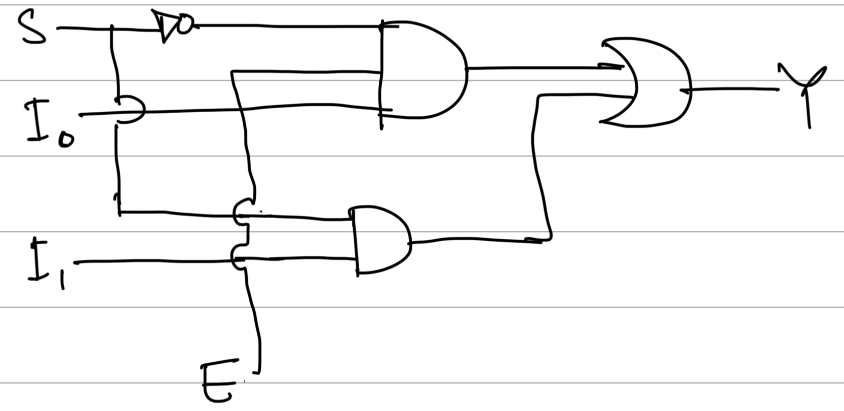
$\underbrace{2:1}_{1}$, $\underbrace{4:1}_{2}$, $\underbrace{8:1}_{3}$, $\underbrace{16:1}_{4}$, $\underbrace{32:1}_{5}$

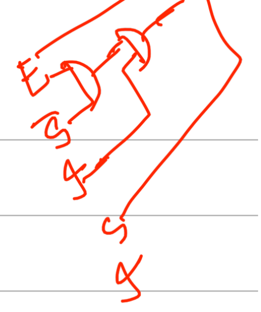
2:1 MUX:-



E	S	Y
0	X	0
1	0	I_0
1	1	I_1

$$Y = E\bar{S}I_0 + ES I_1$$
$$= E(\bar{S}I_0 + SI_1)$$



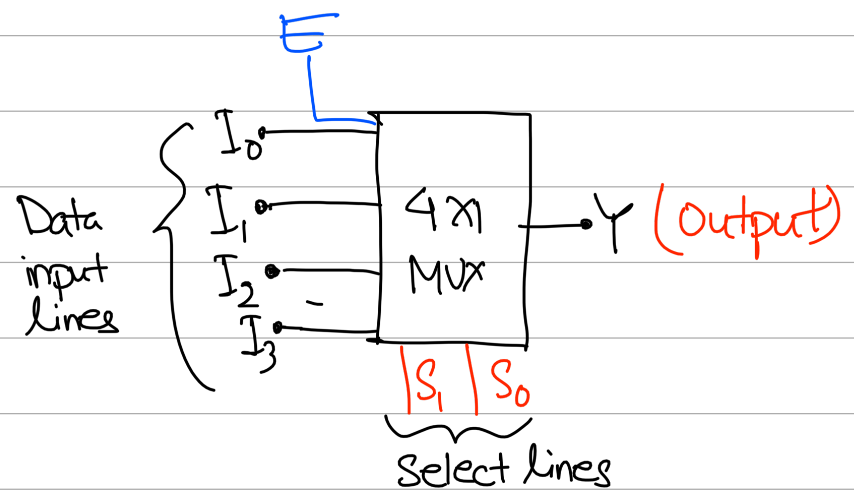


4:1 MUX:-

$n=4$

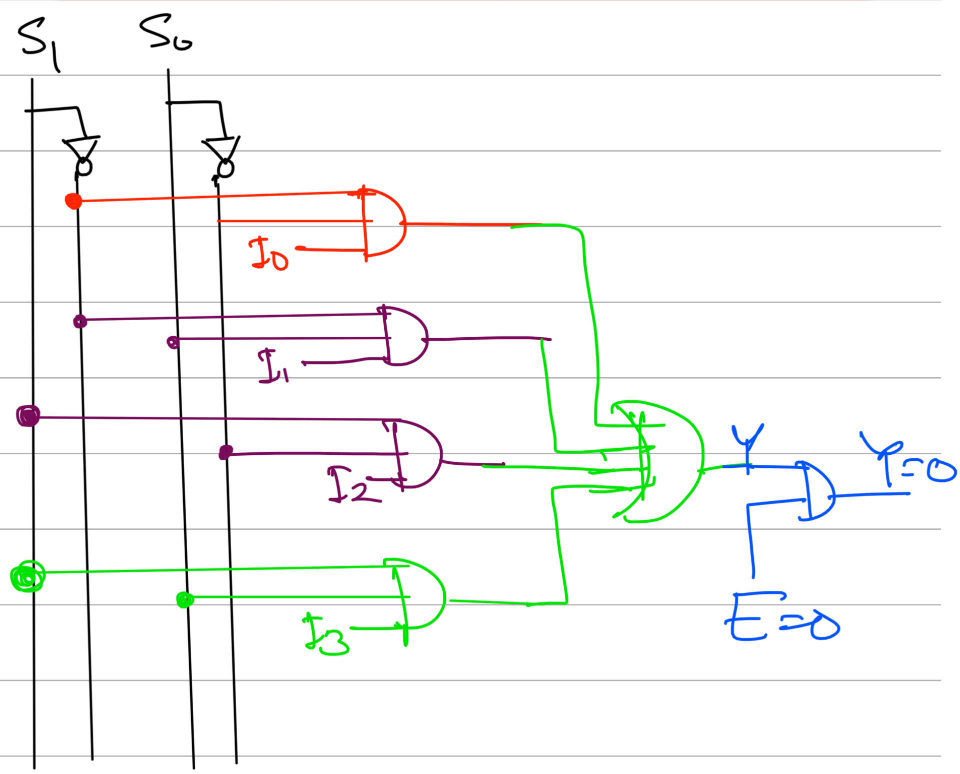
$m = \log_2 n$

$= \log_2 4 = 2$



S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

$$Y = \bar{S_1}\,\bar{S}\,I_0 + \bar{S_1}\,S_0\,I_1 + S_1\,\bar{S_0}\,I_2 + S\,S_0\,I_3$$

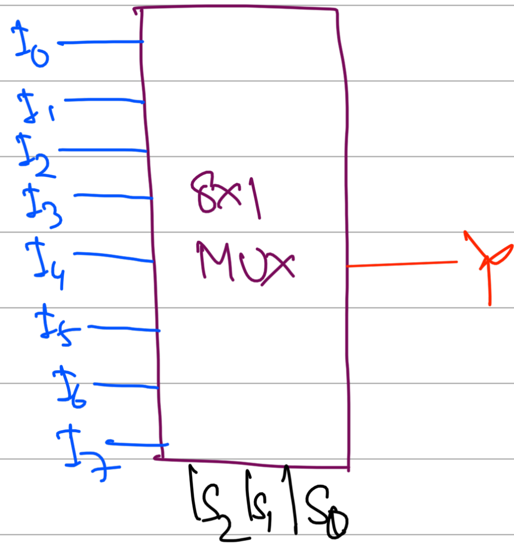


8X1 MUX:-

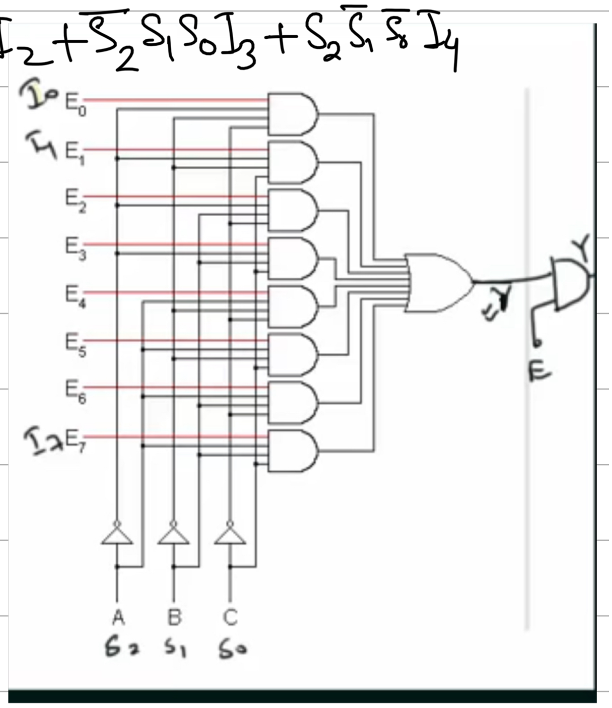
$n=8$

$m=\log_2 8=3$

S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7



$$Y=\bar{S}_2\bar{S}_1\bar{S}_0 I_0+\bar{S}_2\bar{S}_1 S_0 I_1+\bar{S}_2 S_1 S_0 I_2+\bar{S}_2 S_1 S_0 I_3+S_2\bar{S}_1\bar{S}_0 I_4$$
$$+S_2\bar{S}_1 S_0 I_5+S_2 S_1\bar{S}_0 I_6+S_2 S_1 S_0 I_7$$

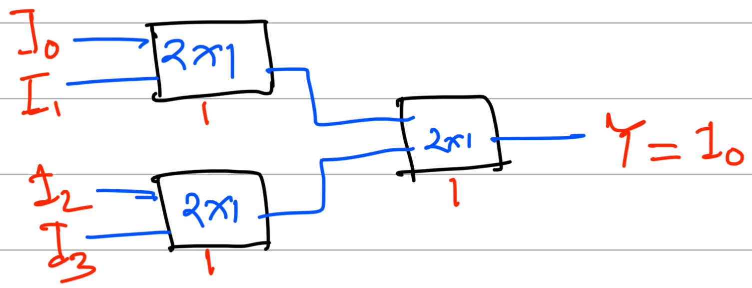


MUX tree Basic | 4x1 MUX using 2x1 MUX

$\frac{4}{2}=2$

$+$ 3 → 2x1 MUX

$\frac{2}{1}=1$



S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

(4) 8x1 MUX By 2x1 :-

S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

$n_1 = 8$

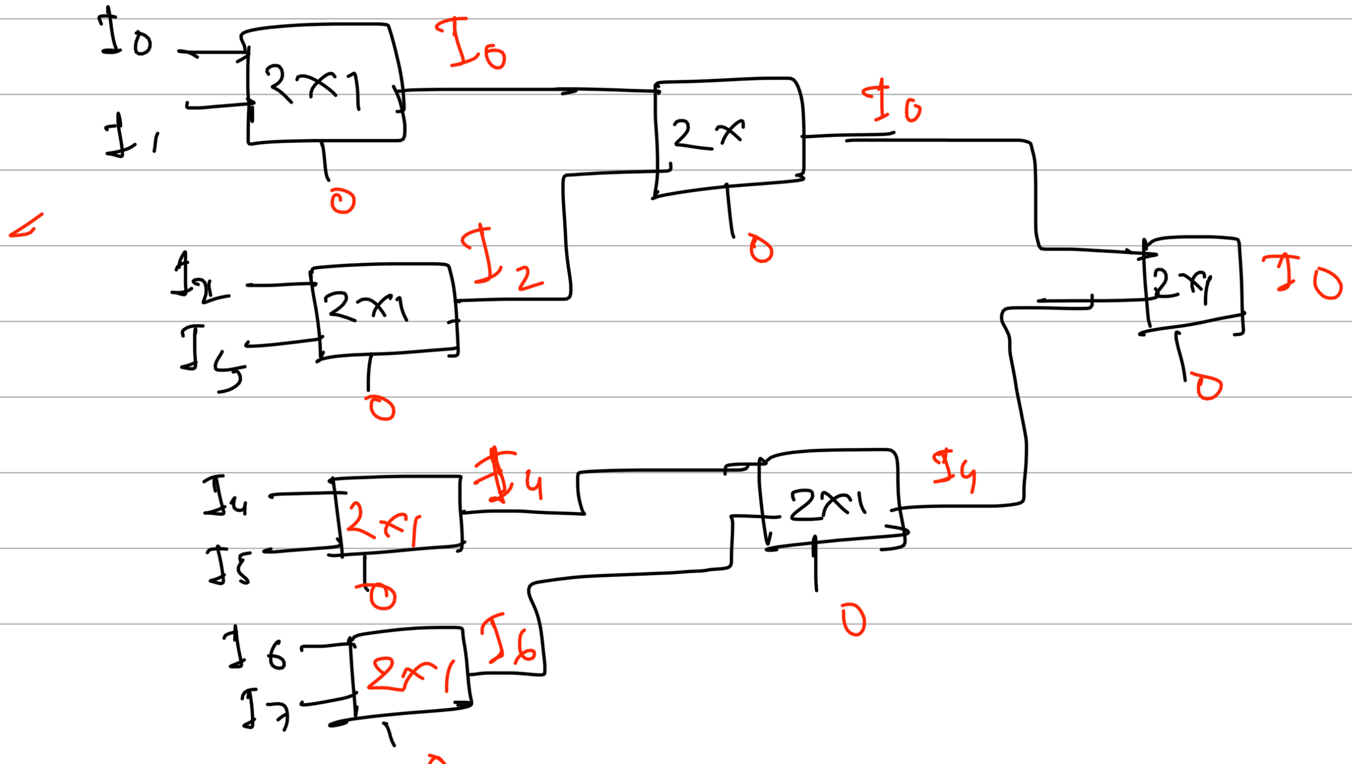
$n_2 = 2$

$\frac{8}{2} = 4$

$\frac{4}{2} = 2$

$2/2 = 1$

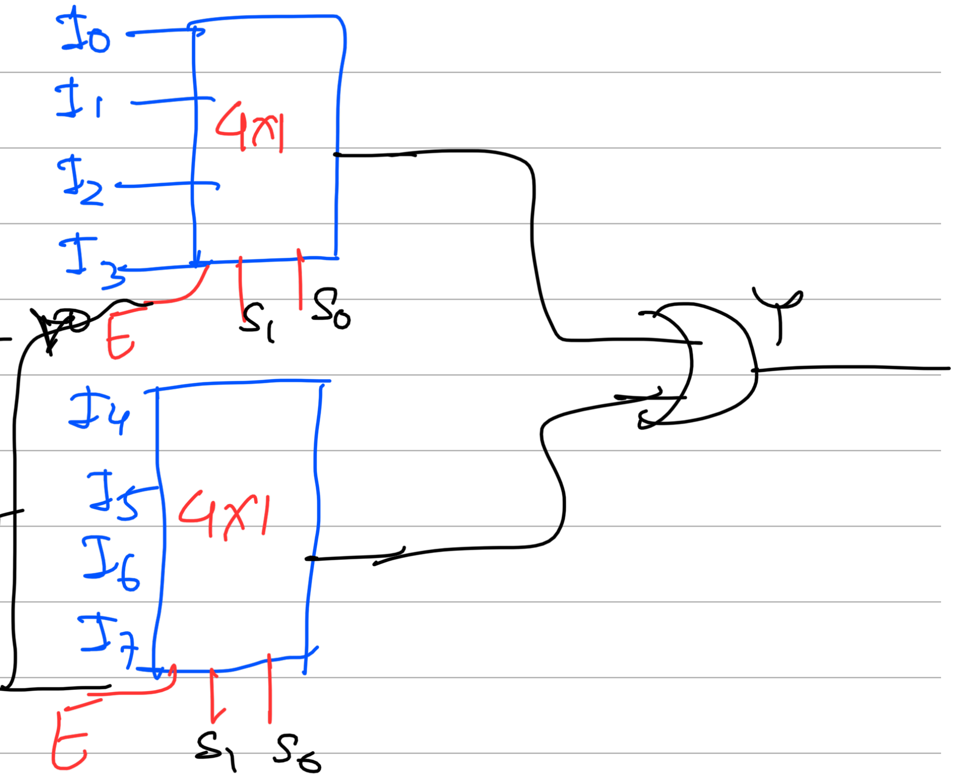
$\boxed{= 7}$



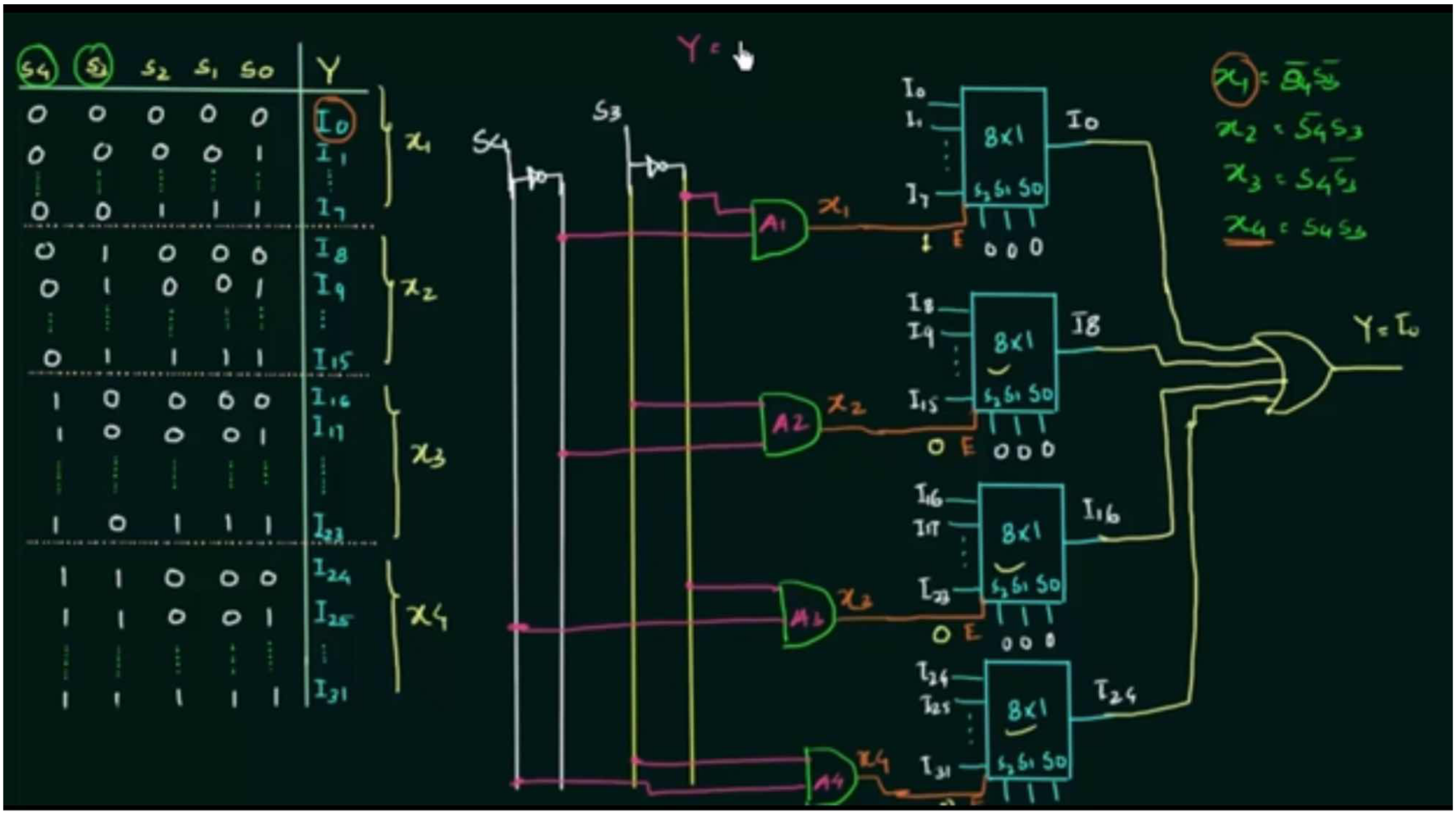
8X1 MUX USING 4X1 (SP) :

using enable can imp.

S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7



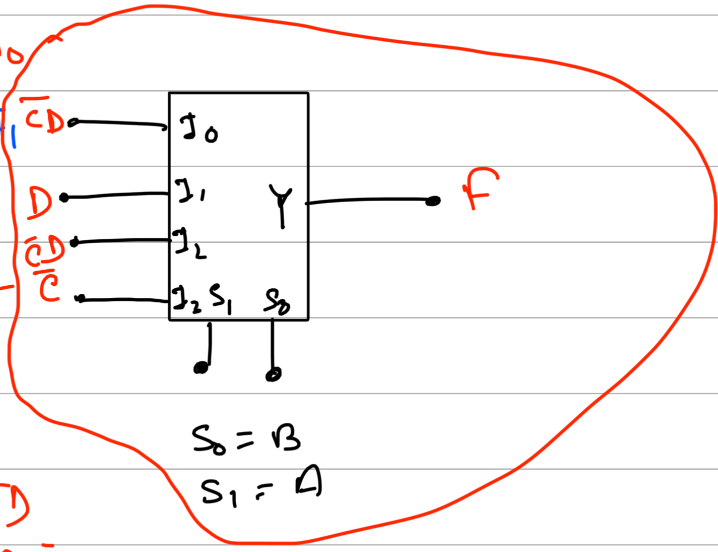
32X1 USING 8X1



Implementation of Boolean function using MUX

Implement:- $F(A,B,C,D)=\sum m(1,4,5,7,9,12,13)$ using MUX(4x1)

AB \ CD	00	01	11	10	
00		1			$\bar{C}D = I_0$
01	1	1	1		$\bar{C}+D = I_1$
11	1	1			$\bar{C} = I_3$
10		1			$\bar{C}D = I_2$



A S_1	B S_0	Y
0	0	$I_0 = \bar{C}D$
0	1	$I_1 = D+\bar{C}$
1	0	$I_2 = \bar{C}D$
1	1	$I_3 = \bar{C}$

1 bit FA using 4x1 MUX :-

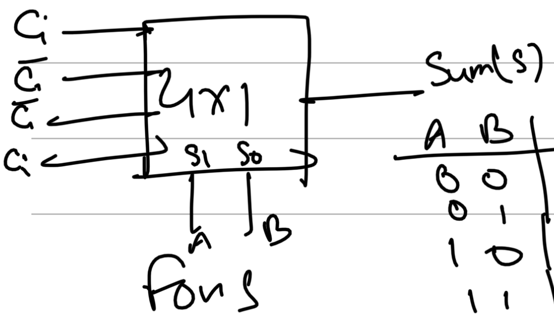
A	B	C_{in}	S	C_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

for S → A=0 → B=0

A \ BC_i	00	01	11	10
0		1		1
1	1		1	

for S

A \ BC_i	00	01	11	10
0				
1				



A	B	S
0	0	$I_0 = C_i$
0	1	$I_1 = \bar{C}_i$
1	0	$I_2 = \bar{C}_i$
1	1	$I_3 = C_i$